

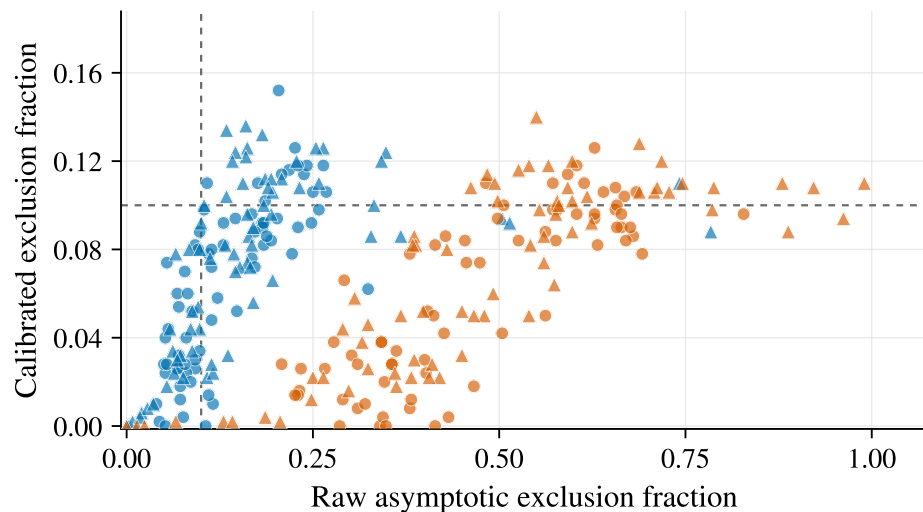
Independent validation of 90% CL_s exclusion

Signal injection $A_{\text{true}} = 5\sigma_{\text{ref}}$; paired background treatments

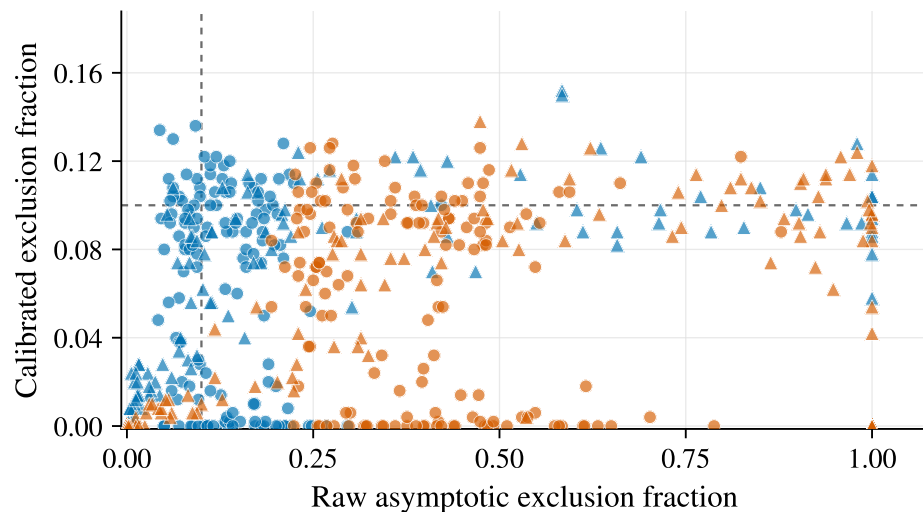
— Gaussian profile — Fixed GP mean ● GP mean truth ▲ Archived stress truth

Complete mass grid: 456 of 456 mass hypotheses; 500 independent spectra per cell

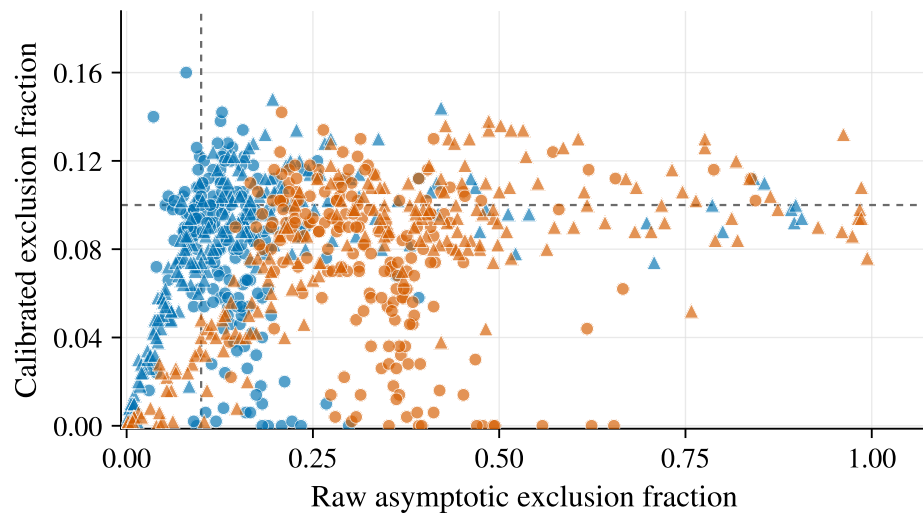
2015, full sample



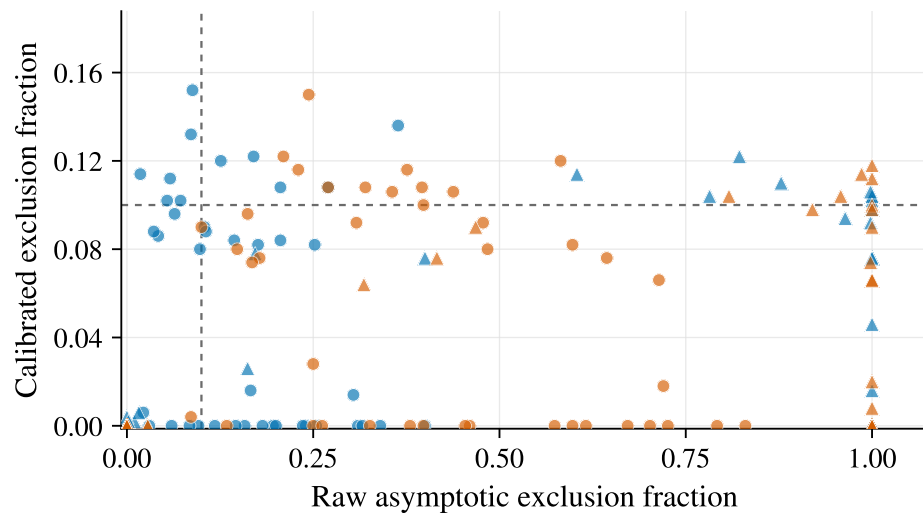
2016, full sample



2021, 10% sample



All three, shared coupling



One point per mass, truth and method; exclusion means $CL_s(A_{\text{true}}) < 0.10$.

Dashed guides mark 0.10. Cells retain their own counts; the calibrated test uses the two-truth envelope.